	Power dissipated = $V_{r.m.s.}^2 / R = 3.8 / 2.0 = 1.9$ W
25	**Ans: C** Magnetic flux through the transformer follows the same sinusoidal function as that of the alternating supply current, i.e. sine curve. Since the induced e.m.f. in the secondary coil is proportional to the rate of change of magnetic flux through the coil (i.e. $d\Phi/dt$), hence the voltage and current at the secondary coil will be a cosine curve. Since $P = I^2R = V^2/R$, the power-time graph will be a cosine-square curve.
26	**Ans: B**